

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2023-2024

Comprehensive Examination
(EC-3 Regular)

Course No. : CCZG502
Course Title : Cloud Infrastructure and Systems Software
Nature of Exam : Open Book
Weightage : 40%
Duration : 2 Hours 30 Minutes
Date of Exam : 18/05/2024 (FN)

No. of Pages	= 2
No. of Questions	= 8

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. Each answer should start from a fresh page.
3. Assumptions made, if any, should be stated clearly at the beginning of your answer.
4. If required to draw any image, then paper & pen can be used, and its photo can be attached.

- Q.1. Consider the factors that the data center plans to implement modular growth, the maintenance team has limited personnel, requiring a design that minimizes ongoing management overhead, cost is a critical factor, and data center's initial budget is tight. Suggest which switch location is preferable in such case with proper justification. **[3 Marks]**
- Q.2. Differentiate deadlock prevention and deadlock avoidance. Also give an analogy to correlate it with real life events. **[3 Marks]**
- Q.3. Explain the concept of hot and cold aisle configurations in data centers. In your answer, discuss a specific use case where this approach is beneficial, outlining the problem it addresses, the mechanism by which it improves data center efficiency, and any potential limitations or challenges that might arise in its implementation. **[5 Marks]**
- Q.4. Explain the booting procedure of a virtual machine, specifically focusing on how it the process will start, its interaction with the hypervisor, and the underlying hardware during the booting phase? Use appropriate diagram. **[5 Marks]**
- Q.5. A company provides a SaaS solution to customers, with varying workloads. The concern is to maintain high performance and ensure a smooth user experience, particularly during peak usage times. Describe and justify the benchmarking tools that can be used to assess the performance parameters required for given conditions. **[5 Marks]**
- Q.6. Consider a system comprised of three processes with the following execution times and periods: **[5 Marks]**

Process	Execution Time (C)	Period (T)
P1	1	10
P2	1	2
P3	2	5

Give a rate-monotonic-schedule (RMS) for the processes through Gantt chart and calculate the corresponding processor utilization.

Q.7. Explain four approaches for the implementation of resource management policies. Take an example and justify which approach is preferable over others. **[7 Marks]**

Q.8. A technology company needs to establish a data center and will host critical business applications, customer databases, and big data. It aims for a scalable, efficient, and secure data center that meets the company's current and future needs. Compare using Link Aggregation Control Protocol (LACP) and static Link Aggregation in a data center environment based on the requirements. **[7 Marks]**
